

Medical Image Segmentation and Applications

Lab- P3: Atlas based Segmentation

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1. Introduction and problem definition

Atlas-based segmentation is a widely recognized method in medical imaging. In medical image segmentation, an atlas acts as a foundational reference, consisting of an intensity image (called a template) along with its associated segmented image, known as the atlas labels. This concept underpins the approach of guiding segmentation in target images by aligning them with the template. During this process, the labels from the atlas are mapped onto the target image using the transformation achieved in registration, a method referred to as label propagation.

The primary objective in this experiment is to create a probabilistic atlas for a brain MRI tissue segmentation that involves registration of brain volumes, propagating labels, and generating the atlas. We implemented Elastix and Transformix registration tools for this purpose. The final objective is to develop a framework that enables the extraction of tissue models, with a particular focus on three main tissue types: White Matter (WM), Gray Matter (GM), and Cerebrospinal Fluid (CSF).

2. Algorithm Analysis and Implementation

2.1 Background

Probabilistic atlases are fundamental to multi-atlas segmentation techniques, formed by registering multiple reference images to a shared space and transferring their labels. The fixed image for registration can be selected randomly or based on metrics like error or similarity. The reference image in the probabilistic atlas might be either the fixed image itself or an averaged image derived from the registered intensity images. Atlas labels are generated by merging the propagated labels from all training images, usually using Averaging and Majority Voting. Averaging computes the class-specific average of propagated labels to create a membership probability map for each voxel, indicating the likelihood that it belongs to a particular region where Majority voting assigns each voxel the label that occurs most frequently at its location across the training set images.

Multi-atlas creation involves the major steps namely i) Registration ii) Label Propagation and iii) Atlas Compilation using the given available training images or a selected subset of the images.

For Registration, one reference image is chosen from the dataset and the remaining images are registered to this fixed image so that they align in a common reference space.

For Label Propagation, transformation parameters obtained from registration are applied to the corresponding labels, effectively relocating the labels to the shared registration space.

Finally, for Atlas Compilation, the atlas is built using the registered images, either by selecting the fixed image from registration or by averaging all images in the dataset. Labels are generated by merging the propagated labels from training images, often through methods such as averaging or majority voting.

In this experiment, our objective was to create a probabilistic brain tissue atlas for segmentation from a given training dataset of 15 brain MRI volumes. We used python programming language for implementation including image registration using the Elastix and Transformix functions that are available in SimpleITK (python). Additionally, we also registered the images manually using the command prompt for the initial experimentation.

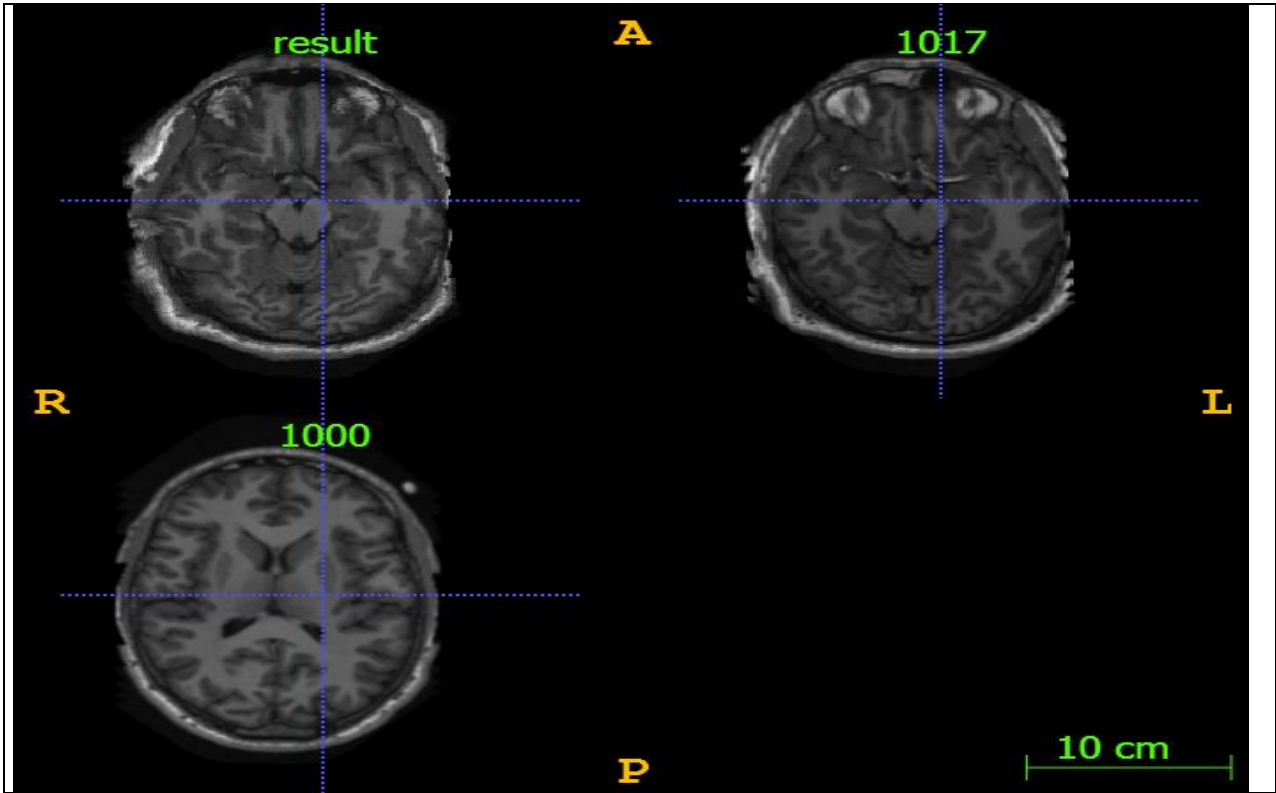
2.2 Implementation (MIRA)

Registration: To register the train images, we needed to select the most optimal fixed image. i.e the fixed image that with the best similarity metric or one that maximizes the segmentation dice coefficients between the fixed and registered labels. To do that, we experimented with all images provided in the train set as fixed images. We also experimented with rigid transformation, affine, B-spline, and combinations of these as provided in the model zoo to determine the most effective models for the registration process and building the atlas. A summary of the experimentation process is provided on the table below.

Parameter File	Best Fixed Image	Average Dice Score
Param 9 Affine	Image 1008	0.6268
Param 9 Elastic	Image 1013	0.45087
Param 9 Affine + Elastic	Image 1017	0.7638
Param 9 Affine + Bspline	Image 1008	0.7221
Param 21 Affine	Image 1008	0.6246
Param 21 Affine + Elastic	Image 1017	0.7621
Param 21 Affine + Bspline	Image 1008	0.7203
Param Default Affine	Image 1013	0.6202
Param Default Bspline	Image 1008	0.7227

Table 1. Summary of Experimentation Process

We therefore proceeded with image 1017 from the training set and the corresponding parameter configuration (Param 9 Affine + Elastic) to perform the final registration and for the rest of our experiments. We saved the registered images as Nifty (.nii) files for use in the atlas generation and for visualization purposes. Example registration results are provided in Figure 1.



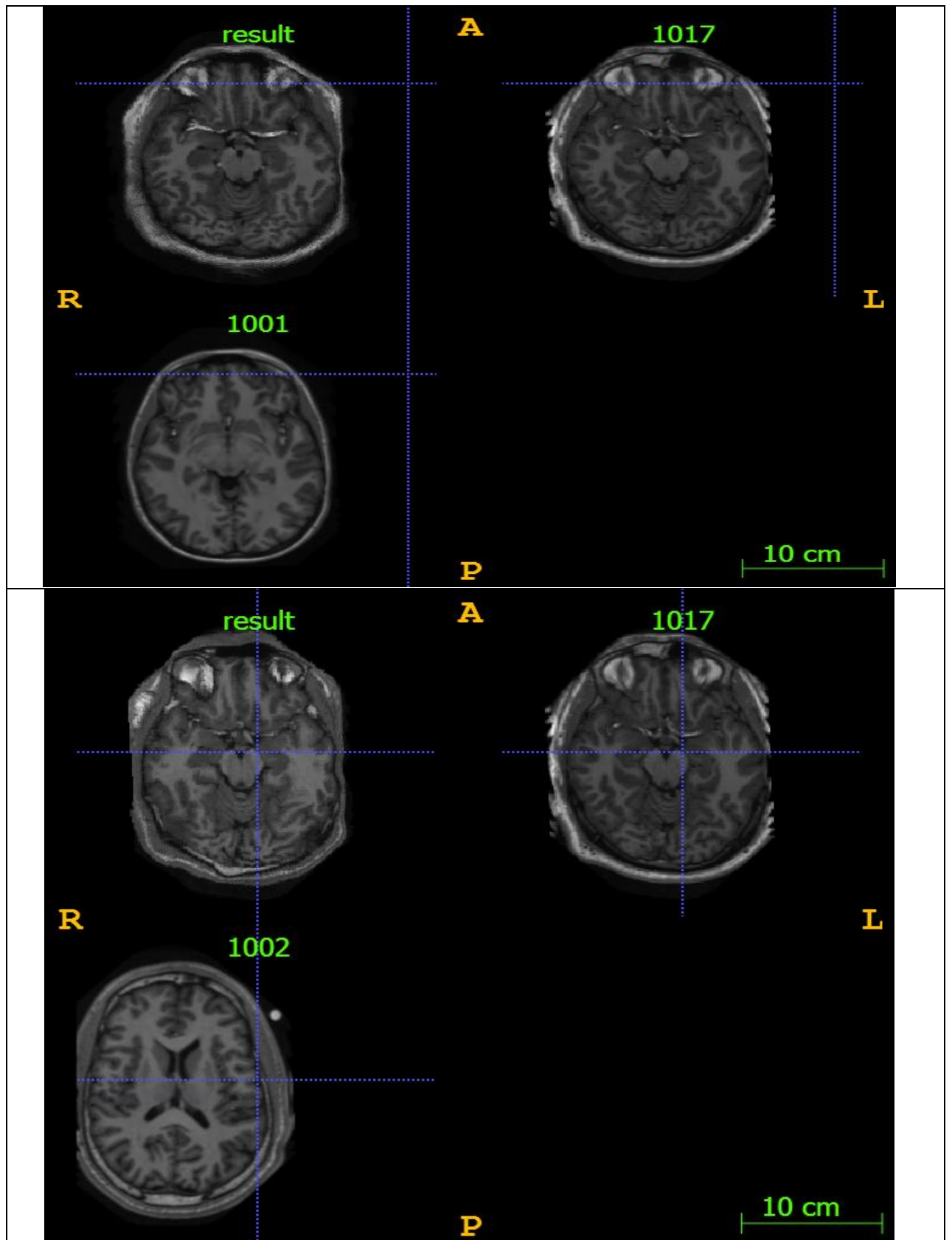
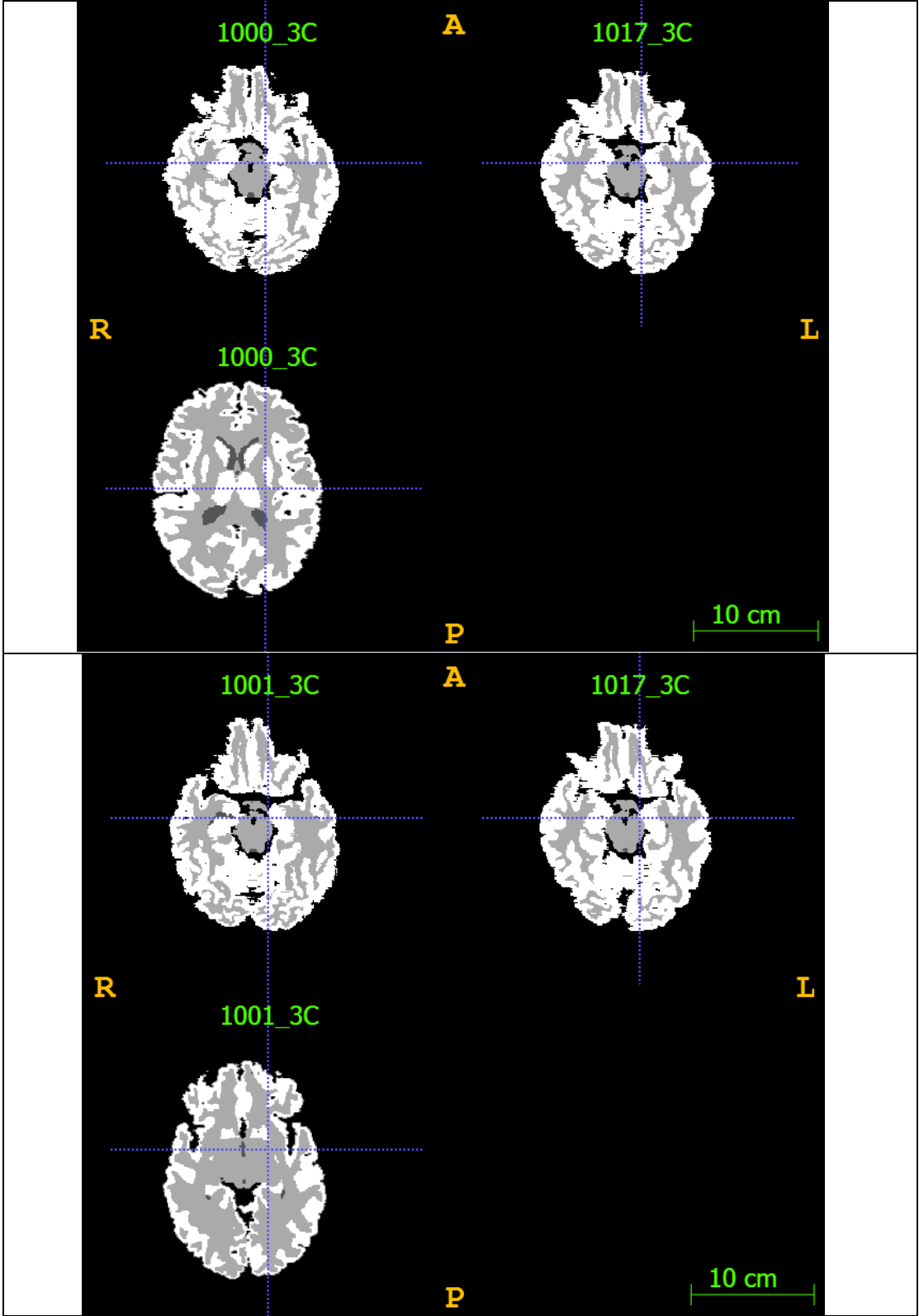


Figure 1. Example registration Results (left: registered image, right: fixed image and bottom: moving image)

Label Propagation: We implemented label propagation in both Python and the command prompt. The registration parameters obtained from aligning the moving images to the fixed image were used to align their

corresponding labels, a crucial step for our probabilistic atlas. In the command prompt, we employed Elastix’s Transformix extension for label propagation, using the “.\transformix.exe” module to apply transformation parameters to the original labels. The parameters were set to save images in Nifty (.nii) format. To streamline the process, we finally used SimpleITK in Python for the entire label propagation of our train set. Example of the propagated labels are provided below:



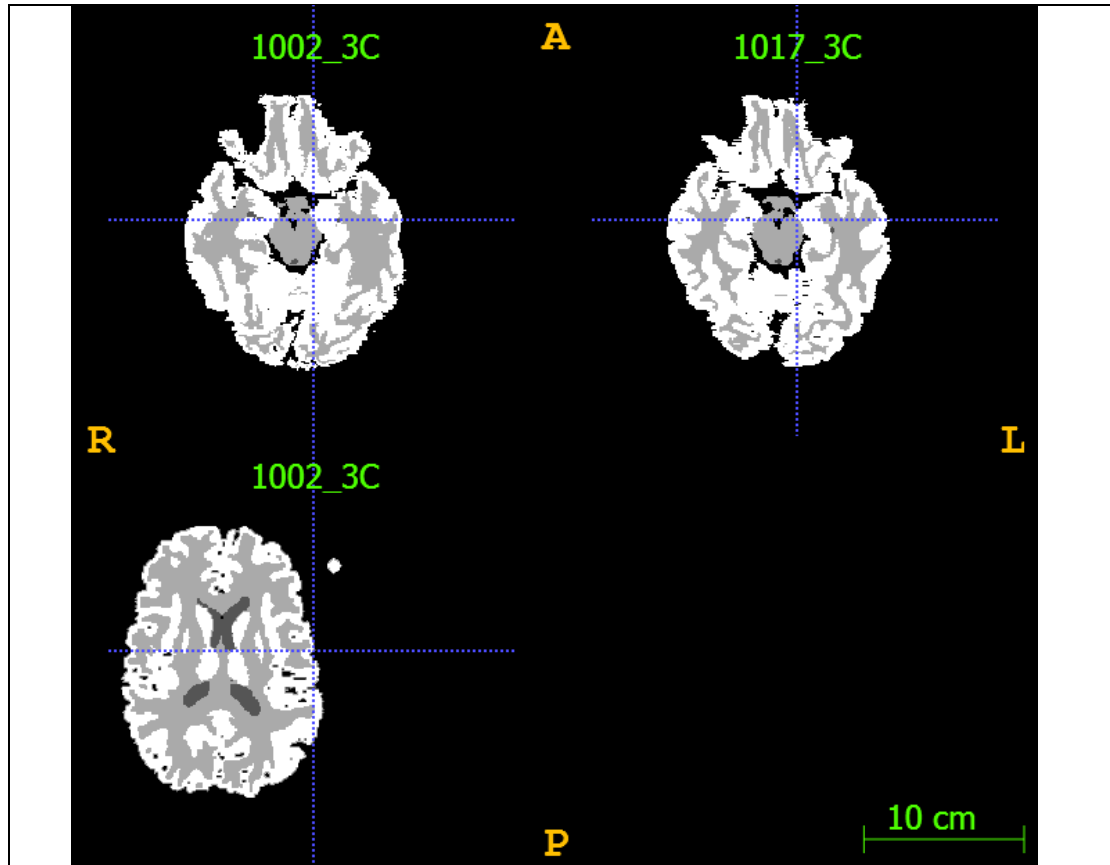


Figure 2. Example Propagated labels (left: registered label, right: fixed label and bottom: moving label)

Atlas Generation: For generating the atlas at first, the template image is created by finding the min-max range of the registered images and averaging them (mean). To build the probabilistic atlas, the propagated categorical labels are split into four binary volumes, one for each class (BG, CSF, WM, GM). The average label for each tissue type is calculated across all subjects, producing membership probability maps. A topological atlas is then derived by applying ‘argmax’ on the averaged probabilistic atlases, assigning each voxel to the tissue with the highest probability at that location in the atlas. Some slices of our generated atlas are shown below:

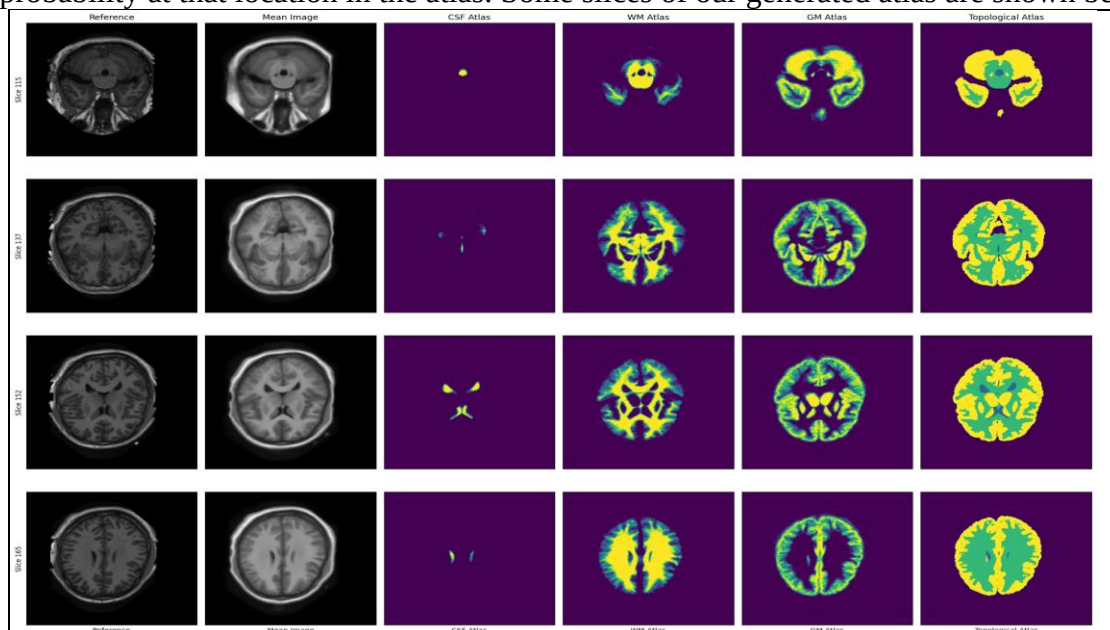


Figure 3. Generated Atlas Example

Tissue Models: One way to implement the tissue models was by using the registered images along with the topological atlas but the easiest way and the one which we used was using the original images and their corresponding labels. For each image and label, the likelihoods for each voxel ($p(x|w_k)$) are computed. These likelihoods represent the conditional probabilities of a voxel's intensity (x) given a tissue class (w_k). Our tissues model is as shown below:

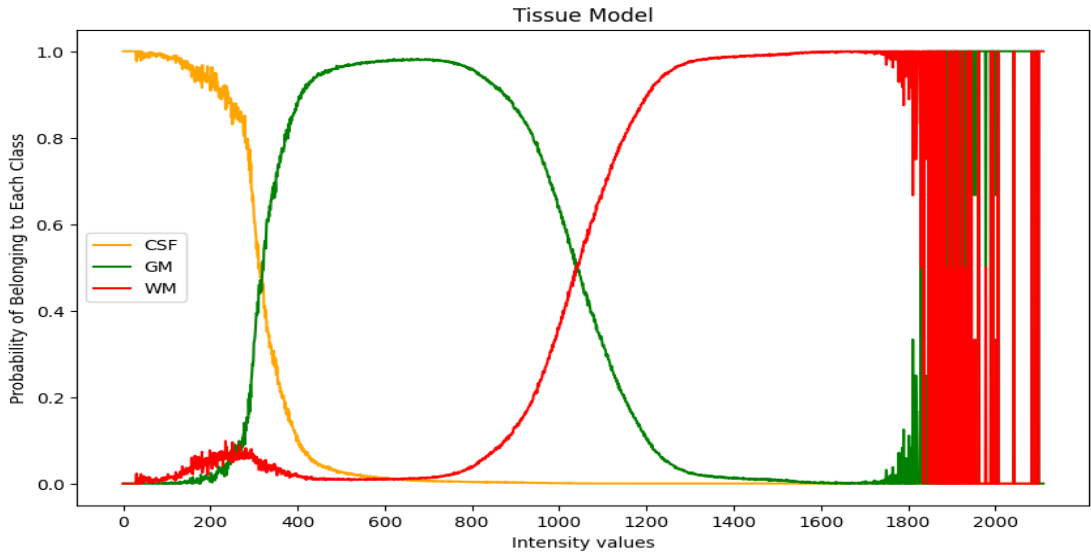
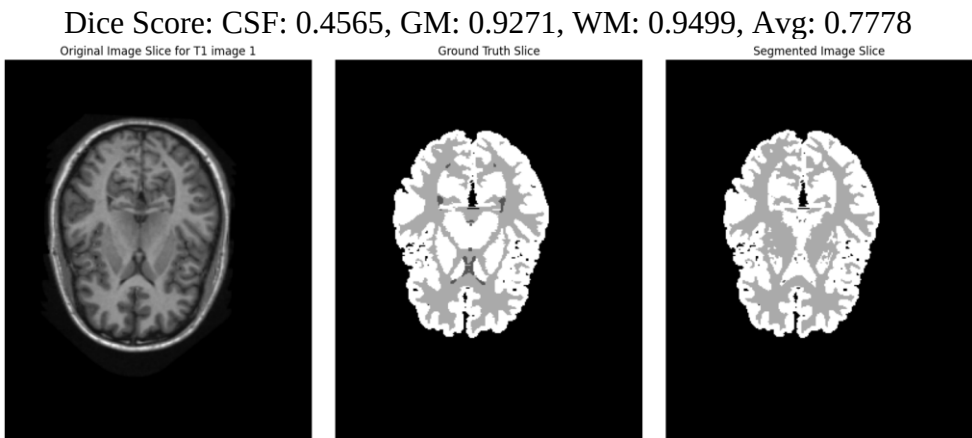


Figure 4. Tissue Model

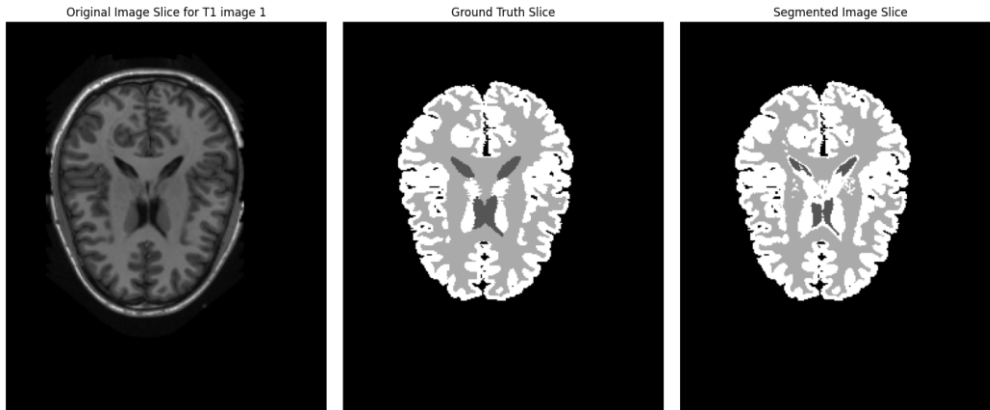
2.3 Implementation (MISA)

For this section of exercise, we used the test set provided to evaluate our implemented segmentation methods. Quantitative and qualitative evaluation of these segmentation methods are provided in the next sections.

Segmentation with Tissue Models: Here, the precomputed tissue model from the train set acts as a look-up-table for our label assignment for the entire test set. Sample outputs are provided below:



Dice Score: CSF: 0.6761, GM: 0.8744, WM: 0.9262, Avg: 0.8256



Dice Score: CSF: 0.8743, GM: 0.9158, WM: 0.9423, Avg: 0.9108

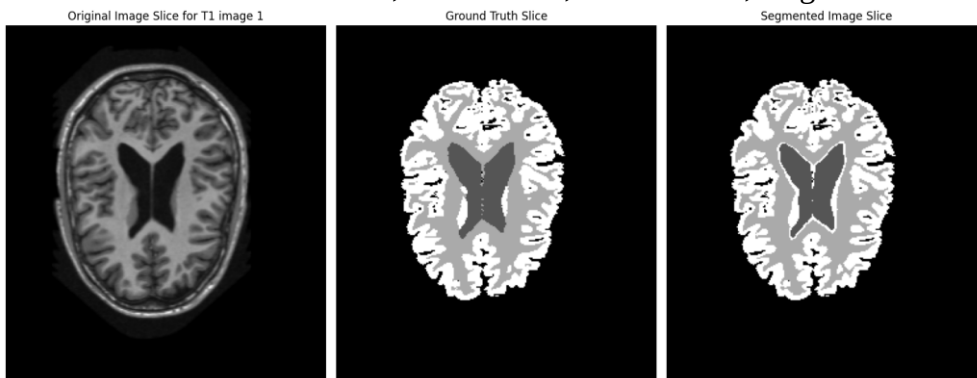
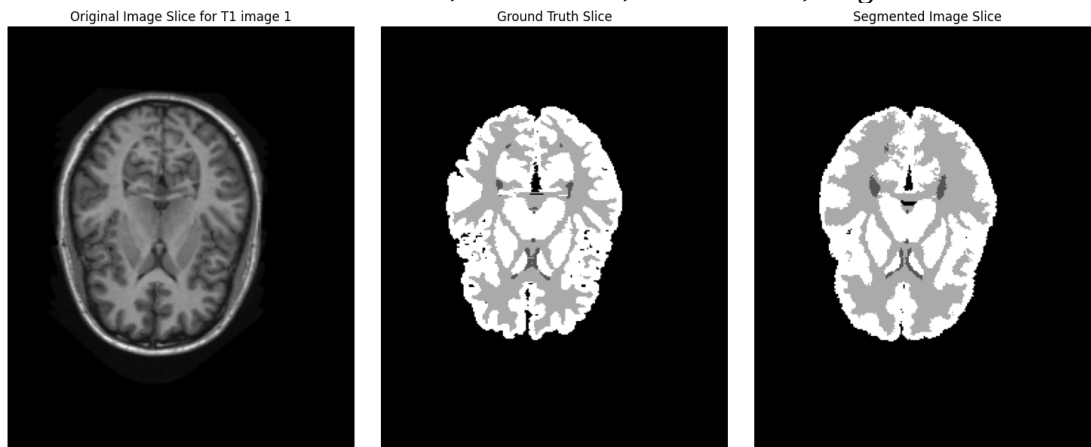


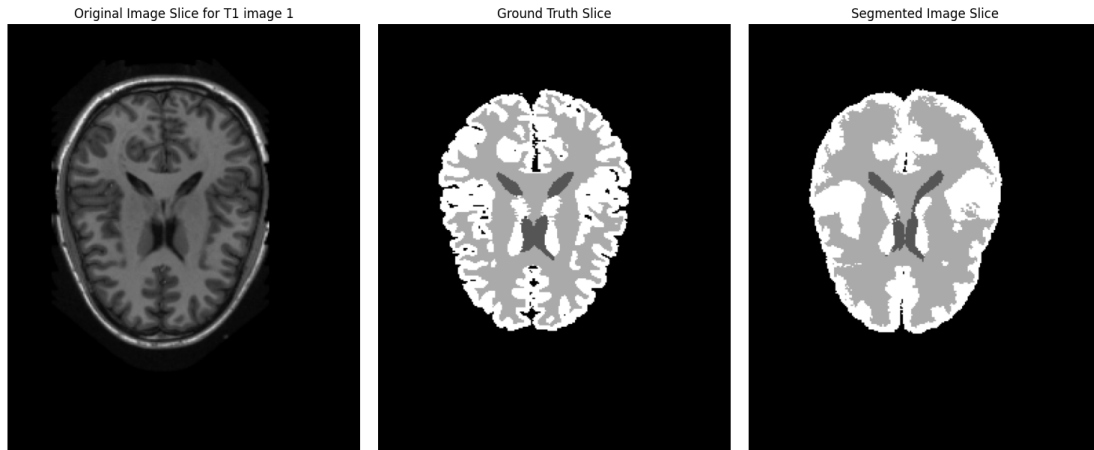
Figure 5. Segmentation with Tissue Model

Segmentation with Label propagation: Here, we used the generated topological atlas to segment our test cases. For each image in the test set, the image is fixed, and the atlas template is registered to the test image. The corresponding transformation parameters are applied to the topological atlas to obtain the segmentation for that test image. The figure below shows some examples of the result obtained.

Dice Score: CSF: 0.6688, GM: 0.7957, WM: 0.7783, Avg: 0.7476



Dice Score: CSF: 0.8230, GM: 0.7937, WM: 0.7618, Avg: 0.7928



Dice Score: CSF: 0.7712, GM: 0.7228, WM: 0.7122, Avg: 0.7354

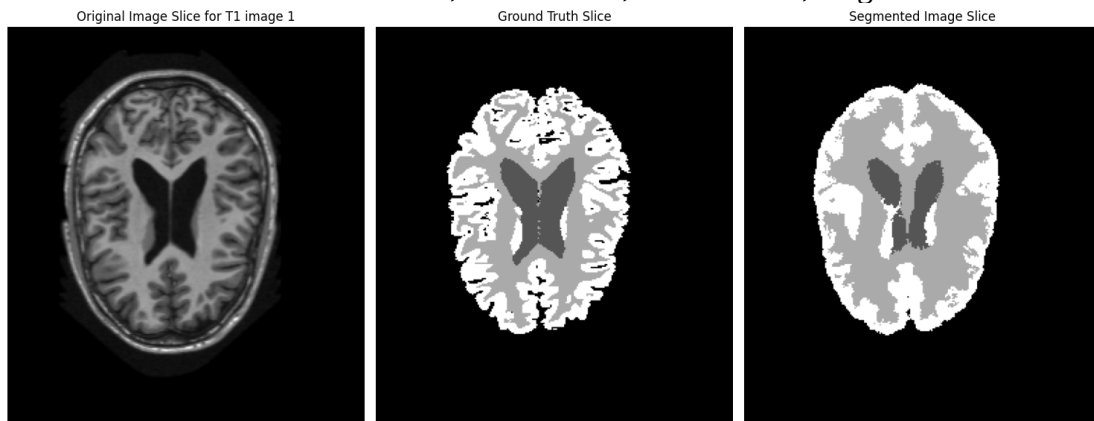
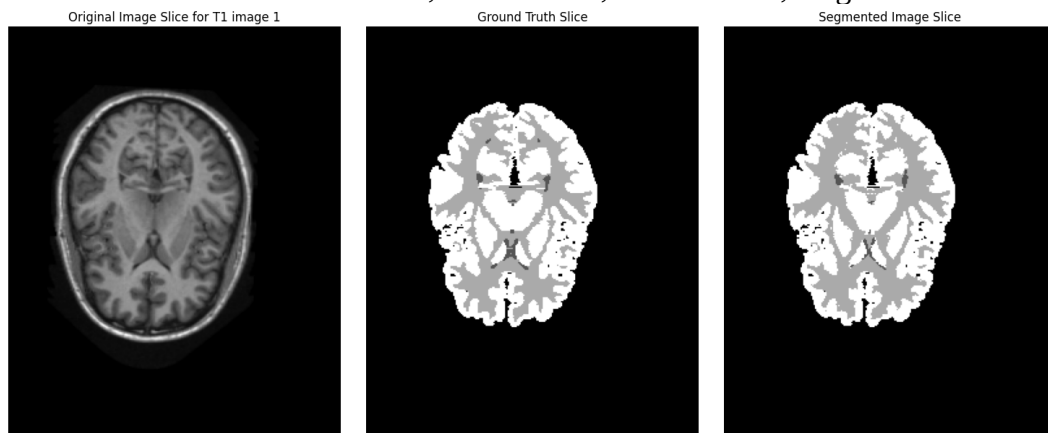


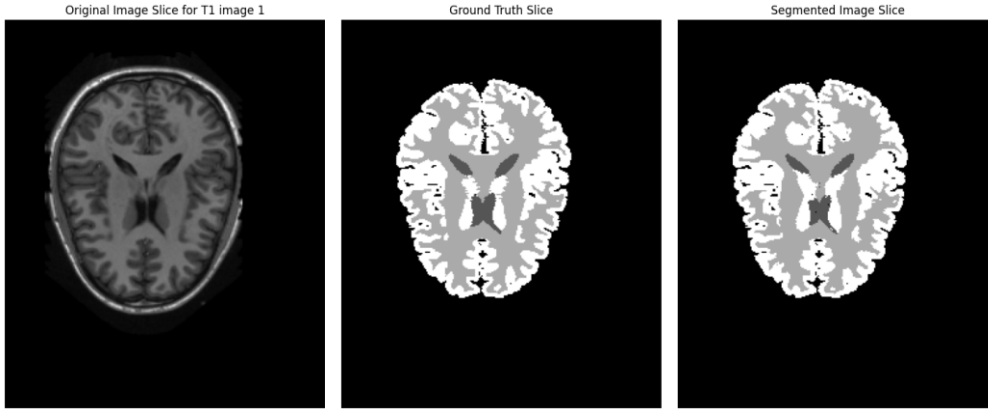
Figure 6. Segmentation with Label Propagation

Segmentation with Tissue Models & Label propagation: Here we combine information from both tissue models and label propagation to segment our test cases. The probabilities of each voxel obtained from the tissue models are weighted by the probabilities from the probabilistic atlas before argmax is used to obtain the final segmentation. We provided some examples of the result obtained.

Dice Score: CSF: 0.7282, GM: 0.9289, WM: 0.9532, Avg: 0.8701



Dice Score: CSF: 0.85524, GM: 0.9218, WM: 0.9494, Avg: 0.9088



Dice Score: CSF: 0.9006, GM: 0.8971, WM: 0.9329, Avg: 0.9102

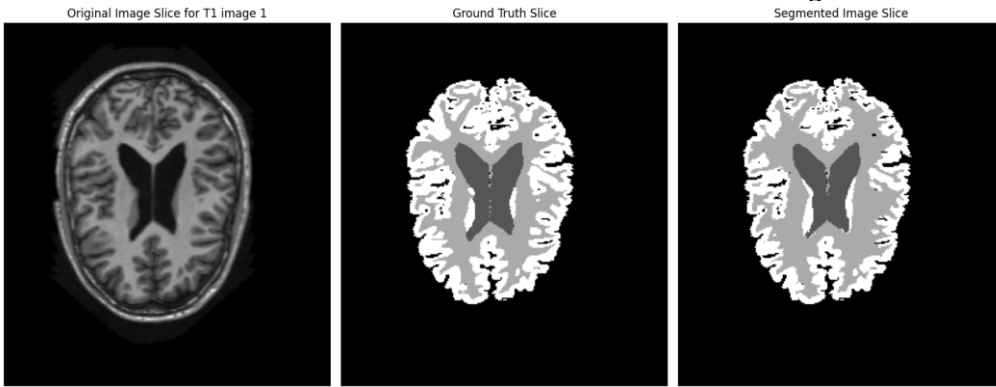
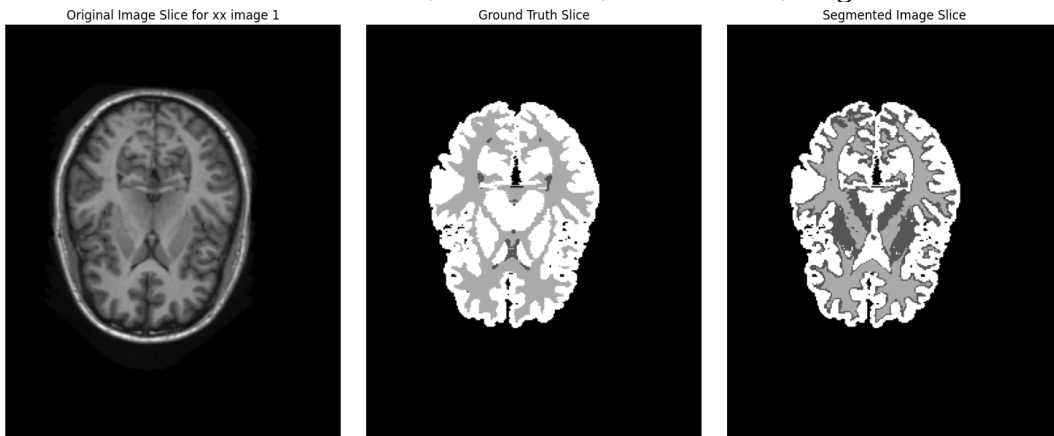


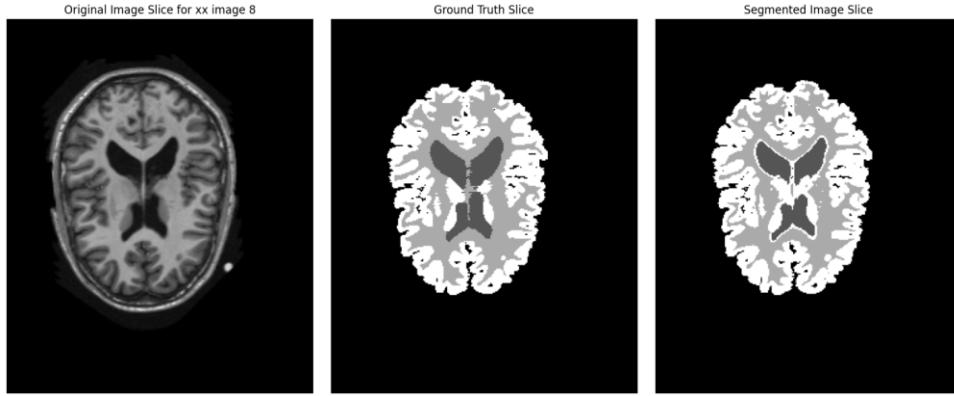
Figure 7. Segmentation with Tissue model and Label Propagation

Segmentation with Expectation maximization and k-means initialization: In one of our previous experiments, we implemented a gaussian mixture model with EM for brain tissue segmentation. In the experiment, we initialized the algorithm with k-means clustering for our initial model parameters (mean and variance). We used the same approach here to segment our test image and obtained the following results:

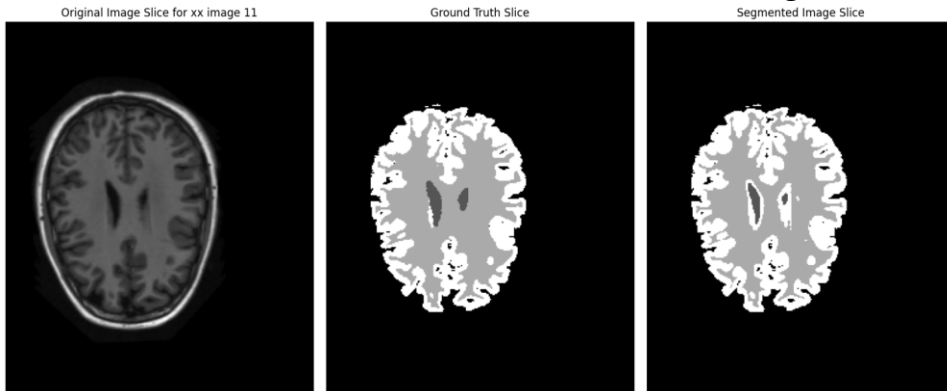
Dice Score: CSF: 0.0015, GM: 0.7617, WM: 0.9109, Avg: 0.5576



Dice Score: CSF: 0.8513, GM: 0.8680, WM: 0.9297 Avg: 0.88301



Dice Score: CSF: 0.7662, GM: 0.8788, WM: 0.9305, Avg: 0.8585



Dice Score: CSF: 0.2587, GM: 0.7783, WM: 0.6321, Avg: 0.5563

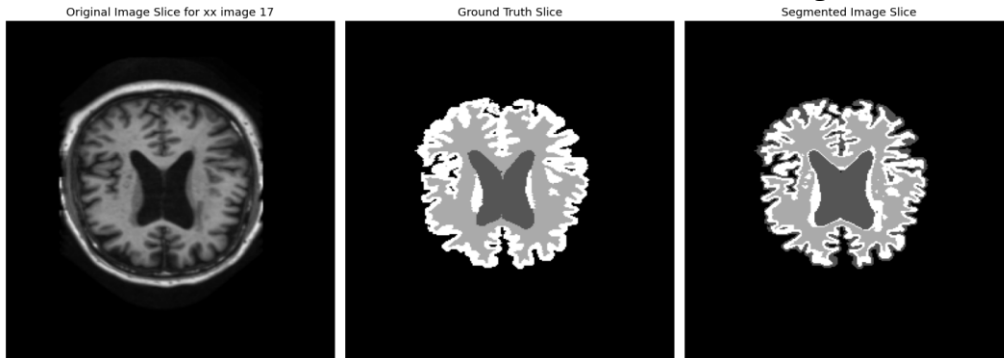


Figure 8. Segmentation with Expectation maximization and k-means initialization

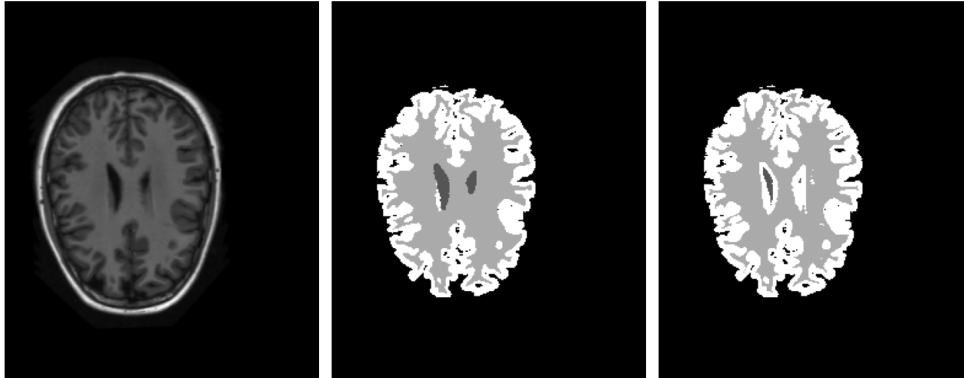
Segmentation with Expectation maximization and Tissue model initialization: Instead of initializing our EM algorithm with precomputed parameters (mean and variance), we use precomputed tissue model (intensity information) as initial weights for our algorithm to estimate the model parameters. This procedure led to the following results.

Dice Score: CSF: 0.5490, GM: 0.8671, WM: 0.9223, Avg: 0.7795

Original Image Slice for xx image 11

Ground Truth Slice

Segmented Image Slice

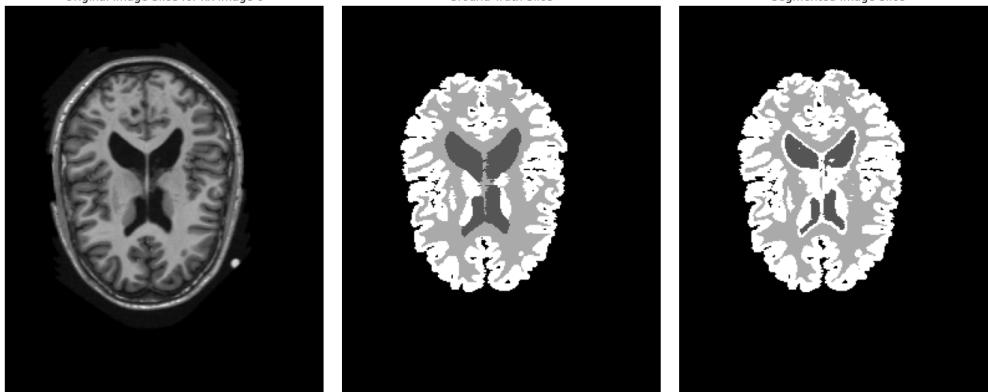


Dice Score: CSF: 0.7315, GM: 0.86202, WM: 0.9222, Avg: 0.8385

Original Image Slice for xx image 8

Ground Truth Slice

Segmented Image Slice



Dice Score: CSF: 0.7487, GM: 0.8674, WM: 0.9212, Avg: 0.8458

Original Image Slice for xx image 3

Ground Truth Slice

Segmented Image Slice

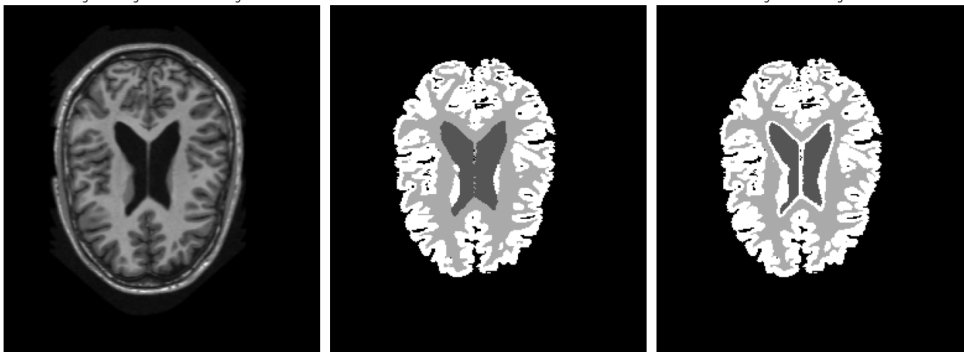
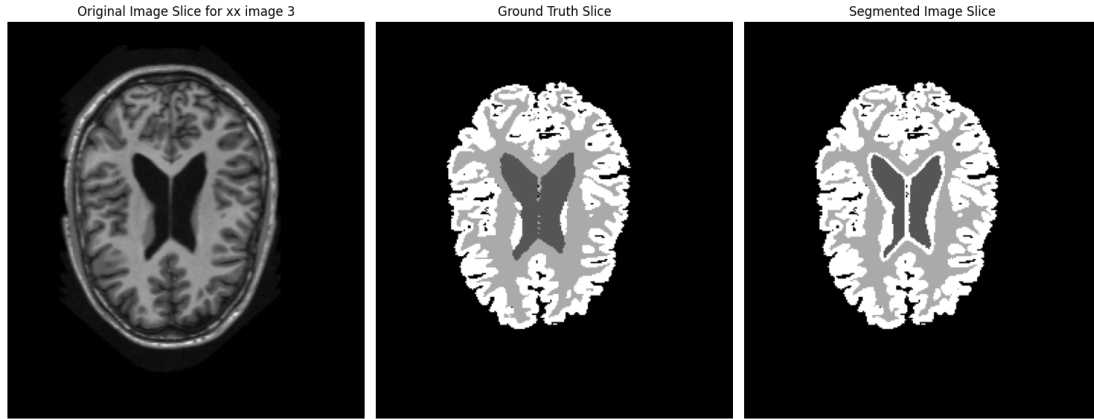


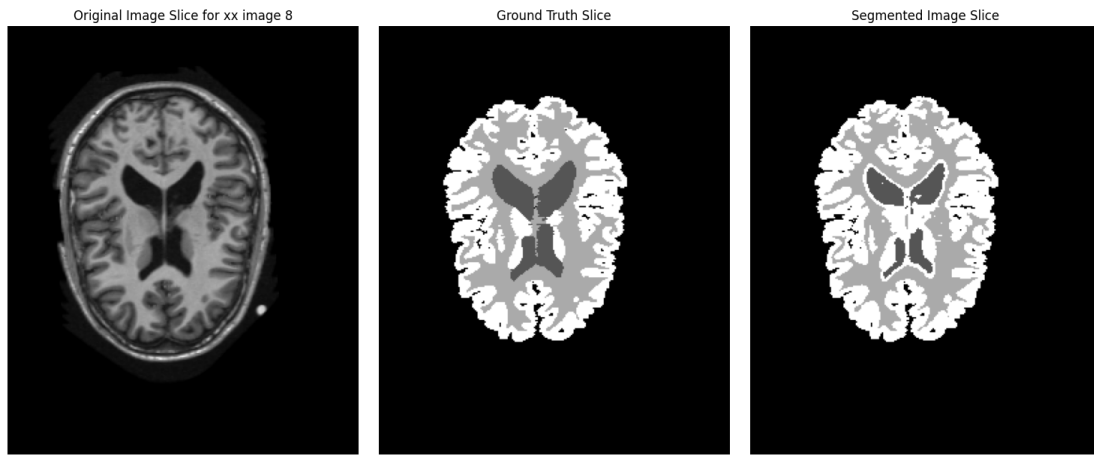
Figure 9. Segmentation with EM and Tissue model initialization

Segmentation with Expectation maximization and label propagation initialization: Here instead of initializing our EM algorithm with precomputed parameters (mean and variance), we use precomputed atlas (position information) as initial weights for our algorithm to estimate the model parameters. This procedure led to the following results.

Dice Score: CSF: 0.7487, GM: 0.8675, WM: 0.9213, Avg: 0.8458



Dice Score: CSF: 0.7315, GM: 0.86202, WM: 0.9222, Avg: 0.8385



Dice Score: CSF: 0.5490, GM: 0.8671, WM: 0.9223, Avg: 0.7795

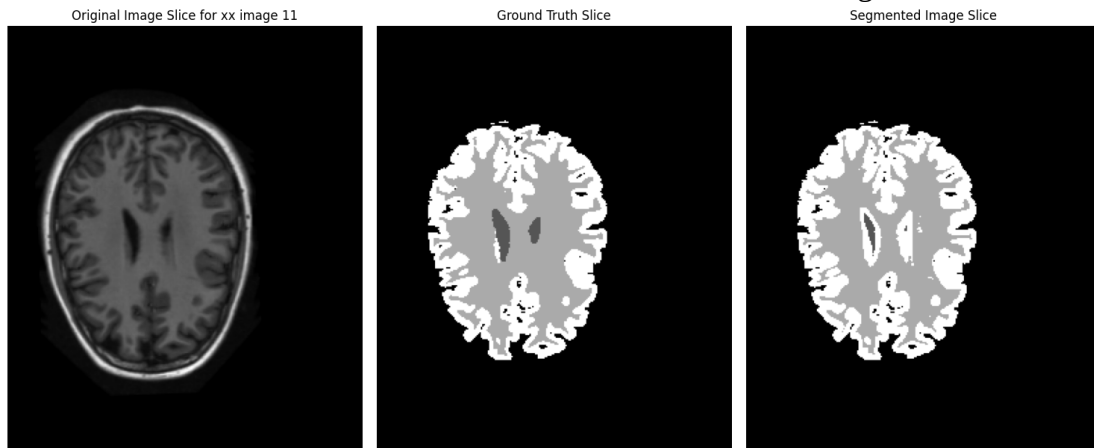


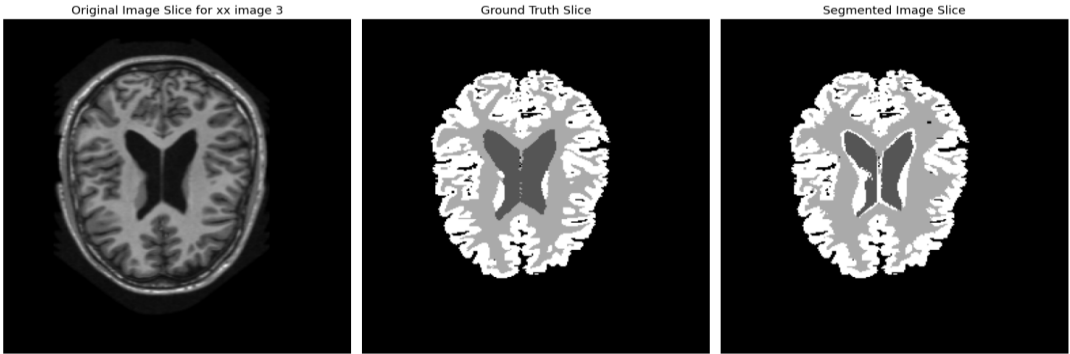
Figure 10. Segmentation with EM and label propagation initialization

Combining EM with Atlas

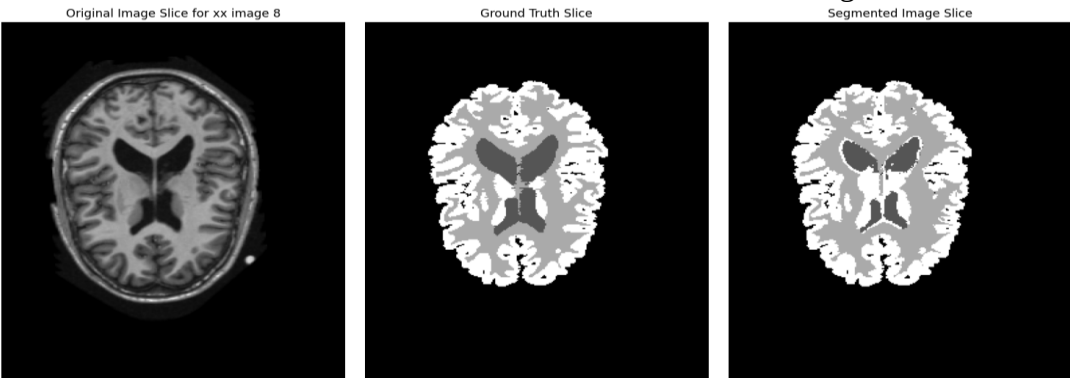
In this section, we compared between our atlas and the well-known MNI atlas in combination with our EM algorithm. For the initialization strategy, we chose to use the initialization with a combination of label propagation and tissue model as it gave the best result in previous sections. In the a posteriori implementation, the atlas information is injected after the EM algorithm has converged while the 'Into EM' section uses an update frequency to inject the contribution of the atlas information.

A posteriori (with our Atlas):

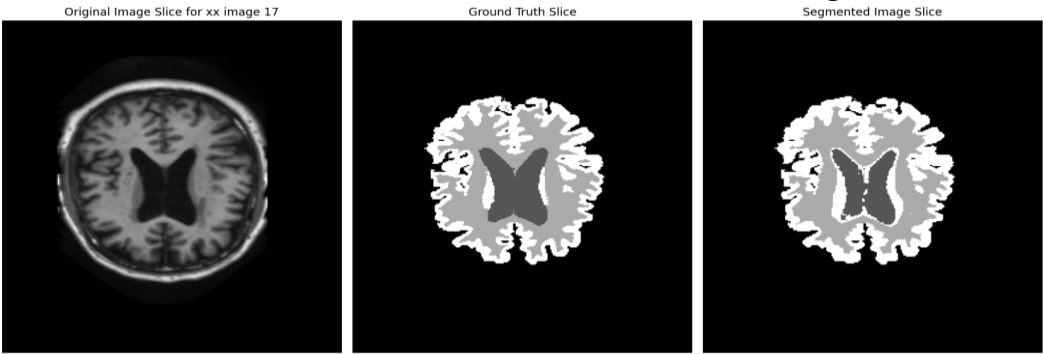
Dice Score: CSF: 0.7742, GM: 0.8789, WM: 0.9221, Avg: 0.8583



Dice Score: CSF: 0.7567, GM: 0.8763, WM: 0.9249, Avg: 0.8526

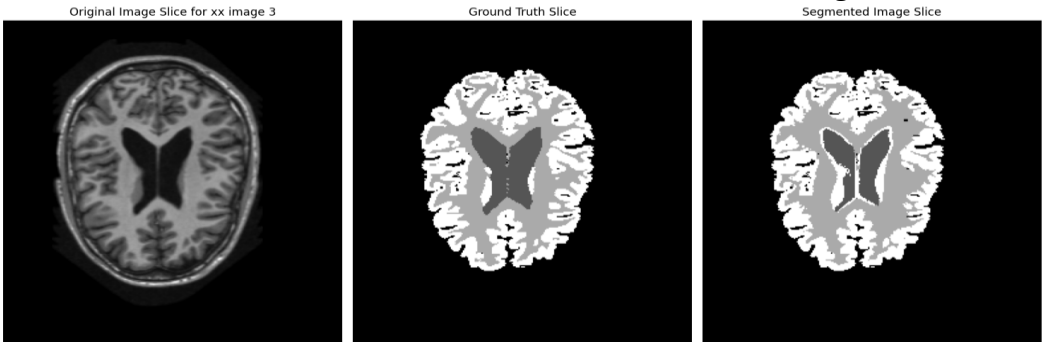


Dice Score: CSF: 0.8151, GM: 0.7991, WM: 0.8596, Avg: 0.8246

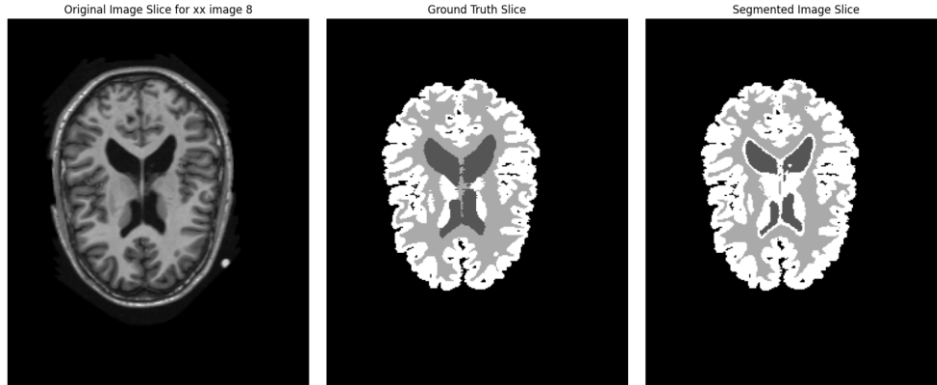


A posteriori (with MNI Atlas):

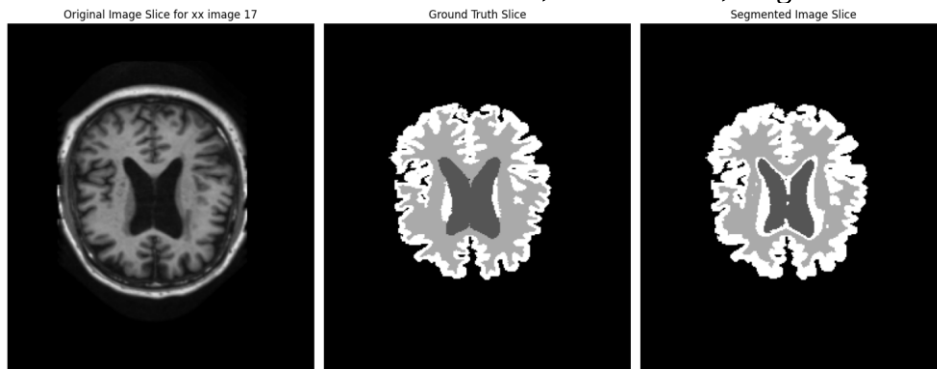
Dice Score: CSF: 0.7541, GM: 0.6557, WM: 0.81105, Avg: 0.7403



Dice Score: CSF: 0.7419, GM: 0.8535, WM: 0.9203, Avg: 0.8386

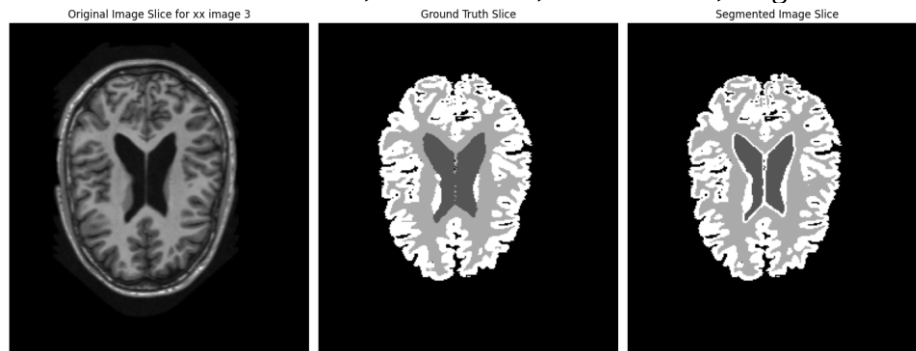


Dice Score: CSF: 0.7002 GM: 0.7904, WM: 0.8836, Avg: 0.7914

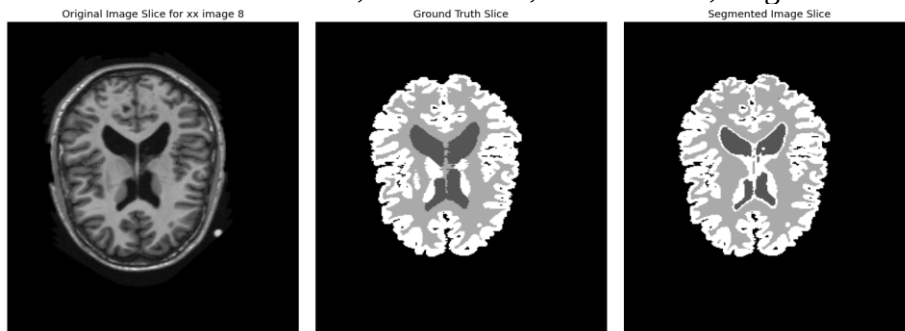


Into EM (with our Atlas):

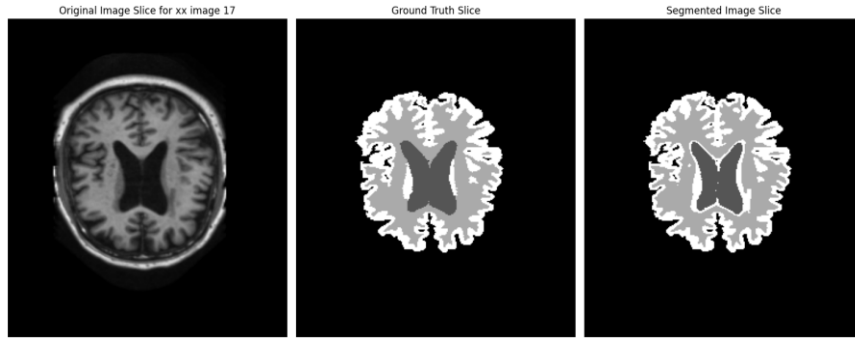
Dice Score: CSF: 0.7933, GM: 0.8978, WM: 0.9227, Avg: 0.8712



Dice Score: CSF: 0.7693, GM: 0.8941, WM: 0.9230, Avg: 0.8621

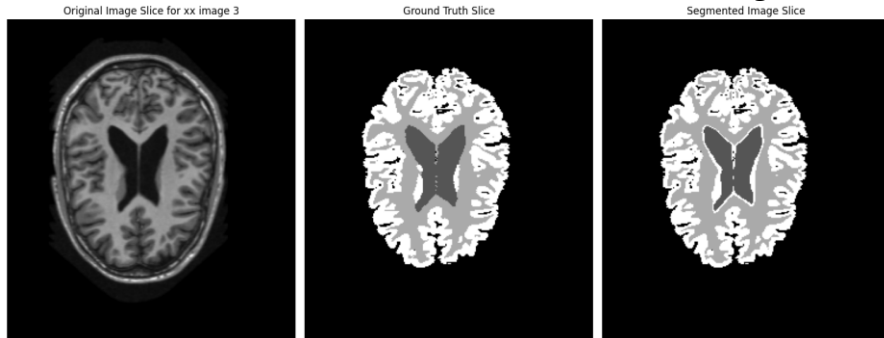


Dice Score: CSF: 0.7720, GM: 0.8924, WM: 0.9124, Avg: 0.8589

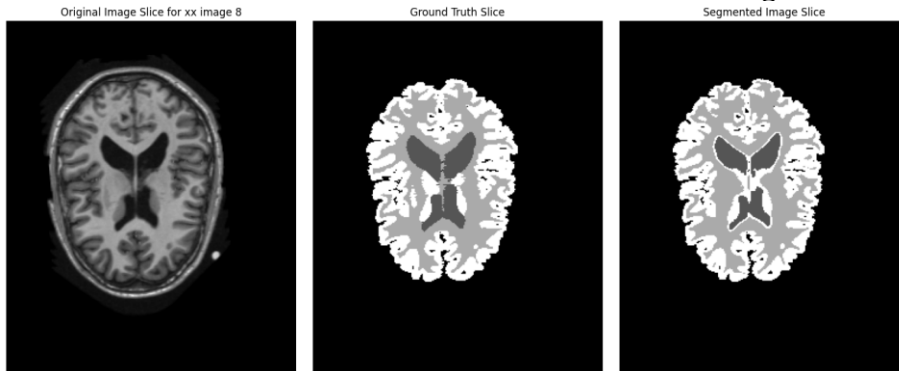


Into EM (with MNI Atlas):

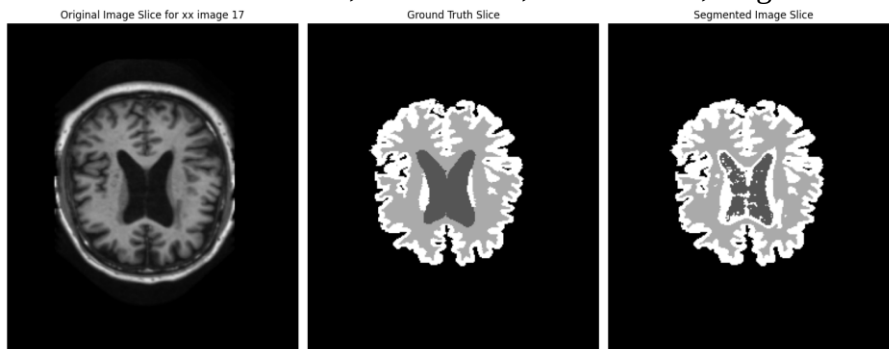
Dice Score: CSF: 0.7889, GM: 0.9191, WM: 0.9418, Avg: 0.8833



Dice Score: CSF: 0.7844, GM: 0.9153, WM: 0.9423, Avg: 0.8807



Dice Score: CSF: 0.6089, GM: 0.8871, WM: 0.9151, Avg: 0.8037



3. Experimental Section and Result Analysis

CSF Segmentation

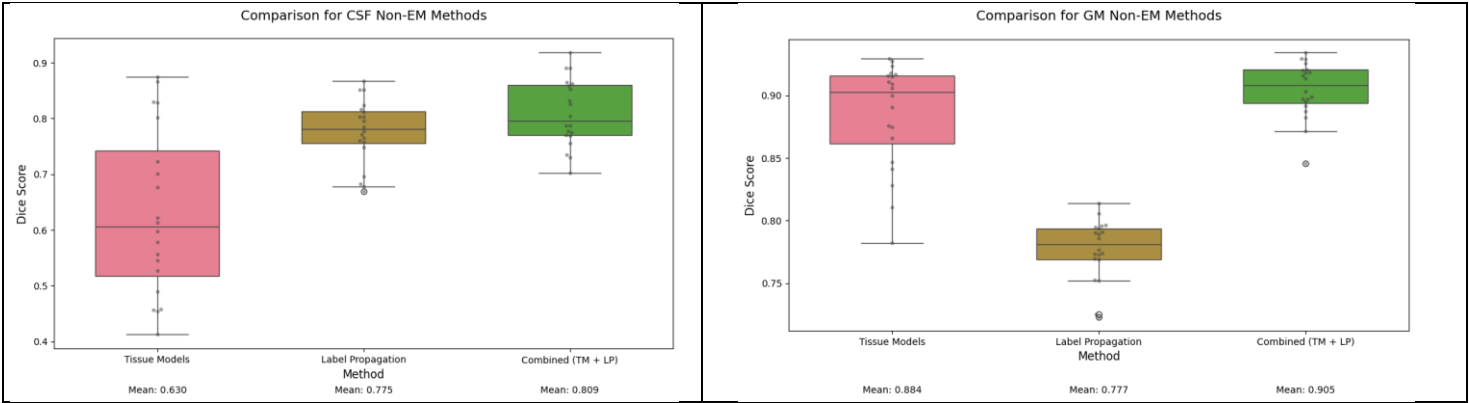
We observed that segmenting the CSF is the most challenging task among the three brain tissues. The primary difficulty arises because the intensity range of CSF is very narrow and overlaps considerably with that of GM, making it hard for intensity-based models to distinguish between the two. From our analysis, we noticed that in the pure EM methods, all three initialization strategies had severely failed to segment some of the CSF effectively. This likely stems from low variability between CSF and GM intensities, causing the algorithm to struggle with distinguishing these tissues as it's convergence depends on the distinctiveness of the intensity distributions. When we segmented the test cases with non-EM methods, the results were much higher, this could be because they directly leverage the anatomical and spatial information with no learning/training being done, although there was a noticeable difference in the pure tissue model segmentation as it leverages only intensity information.

When we introduced atlases into the segmentation process, we observed a slight improvement in the results, with the most significant enhancement achieved when using our own atlas in combination with label propagation and the combined (Label prop + Tissue model) method. It was also evident that increasing the update frequency significantly improved the performance of the "Into EM" method. Overall, our atlas (a posteriori) approach proved to be the most effective at segmenting the CSF, suggesting that incorporating prior anatomical information can significantly aid in distinguishing CSF from GM.

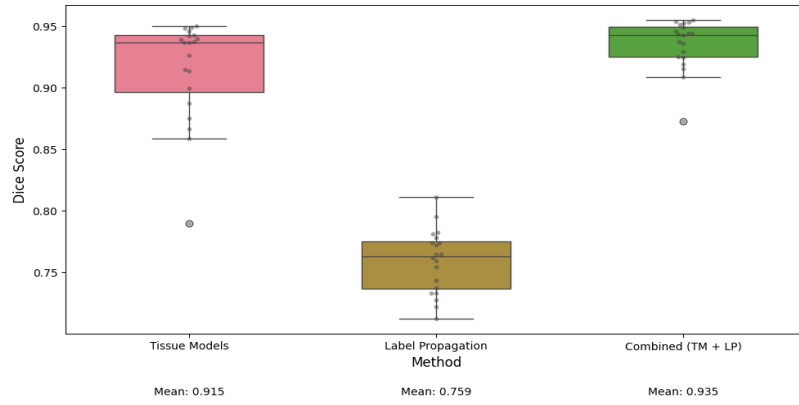
GM and WM Segmentation

In contrast to the challenges faced with CSF segmentation, most of the methods we experimented with produced very good segmentation results for both grey matter (GM) and white matter (WM), with the only exception being the MNI (a posteriori) method. The combination of tissue models and label propagation in the non-EM method yielded the best results for both WM and GM, which is contrary to our observations in the CSF segmentation where these methods struggled.

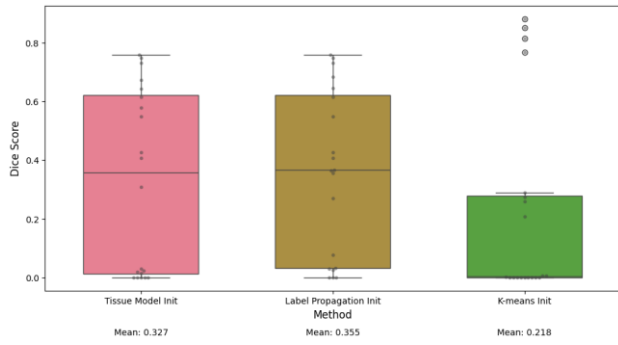
Interestingly, our atlas (a posteriori) method consistently showed performance comparable to the best non-EM method for GM and WM, and it was also the best-performing model in the CSF segmentation. This consistency across all three tissues indicates that our atlas (a posteriori) model is the most robust method we developed. Therefore, we can conclude that our atlas (a posteriori) model is the most effective classic segmentation method among those we implemented, offering reliable performance across all main brain tissues.



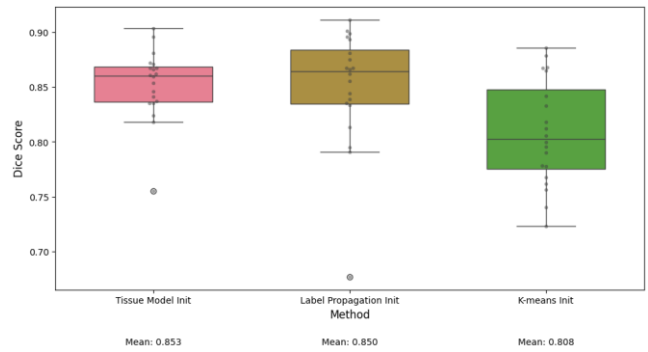
Comparison for WM Non-EM Methods



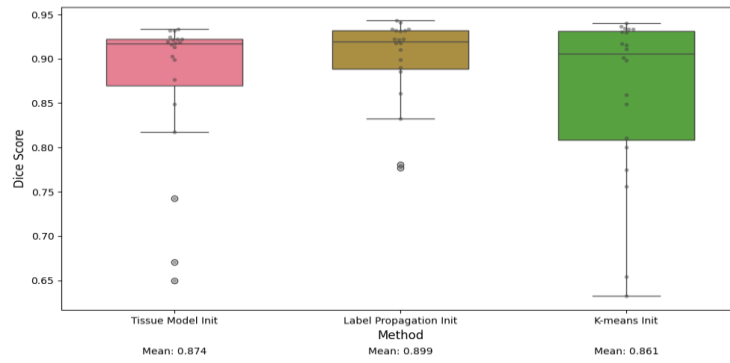
Comparison for CSF with EM Methods



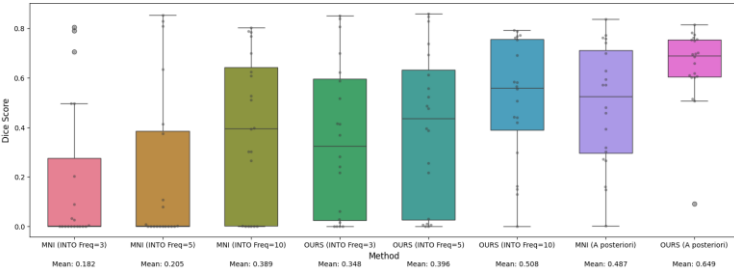
Comparison for GM with EM Methods



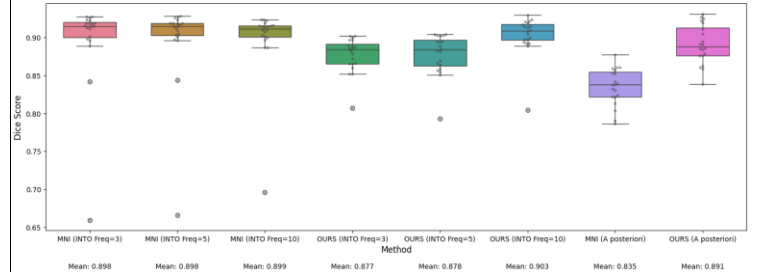
Comparison for WM with EM Methods



Comparison between the MNI and Our Atlas for CSF (Label Prop Only)



Comparison between the MNI and Our Atlas for GM (Label Prop Only)



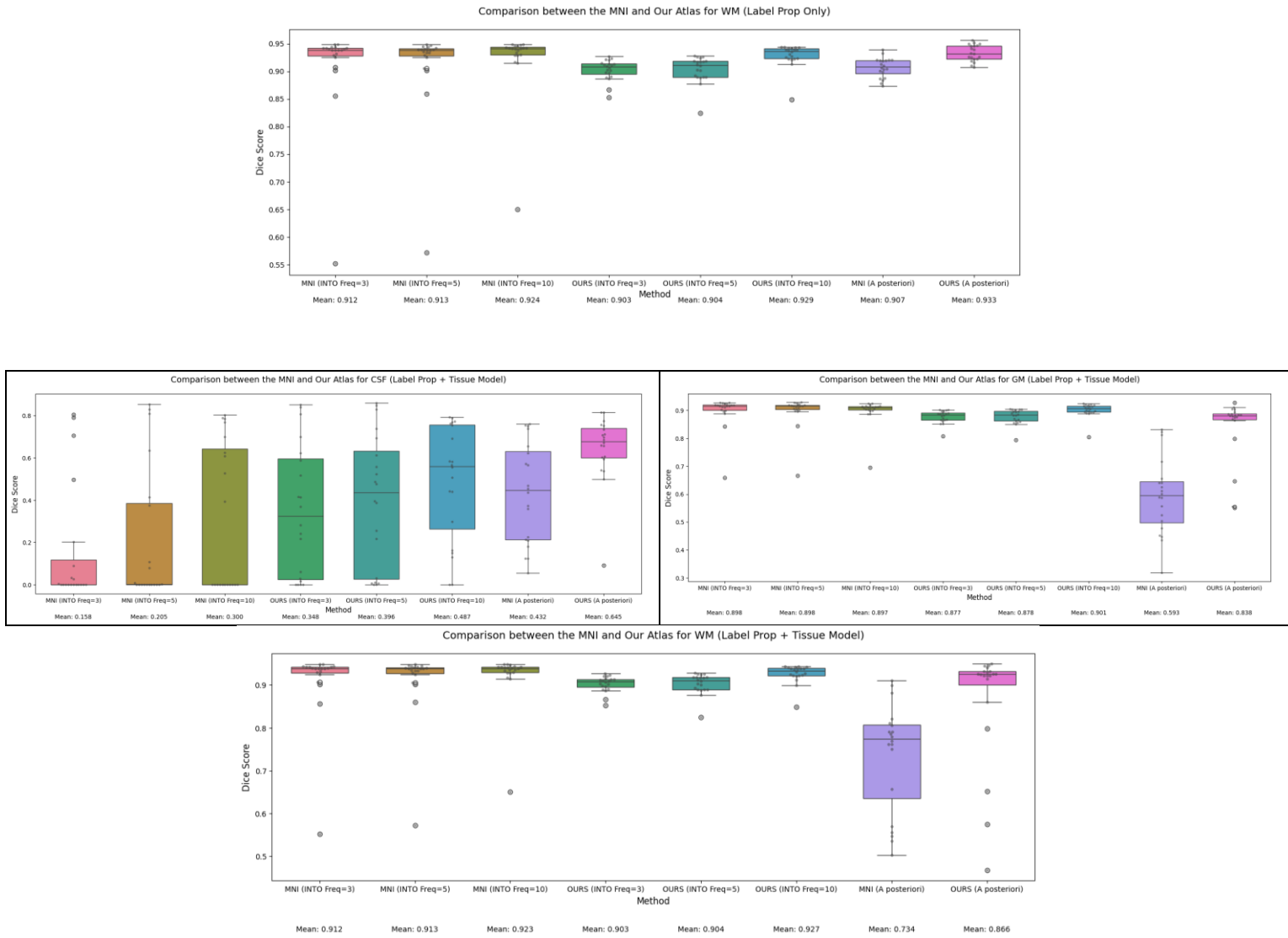


Figure 11. Comparison Box Plots

The table below shows a summary of the segmentation result for each method across all test images.

METHOD	MEAN CSF	MEAN GM	MEAN WM	AVG.
Tissue Models	0.6304	0.8843	0.9148	0.8098
Label Propagation	0.7754	0.7771	0.7589	0.7705
Tissue Model & Label Propagation	0.8105	0.9043	0.9343	0.8830
EM Algorithm				
K-Means Init	0.2182	0.8081	0.8608	0.6291
Tissue Model Init	0.3266	0.8525	0.8740	0.6844
Label Propagation Init	0.3545	0.8499	0.8992	0.7012
Using Label Propagation Only				
A posteriori (our Atlas)	0.6492	0.8913	0.9331	0.8246
A posteriori (MNI Atlas)	0.4823	0.8339	0.9060	0.7408
Into EM (our Atlas, freq=5)	0.3963	0.8776	0.9035	0.7258
Into EM (our Atlas, freq=3)	0.3481	0.8770	0.9027	0.7093
Into EM (our Atlas, freq=10)	0.5081	0.9033	0.9288	0.7801
Into EM (MNI Atlas, freq=5)	0.2162	0.8970	0.9114	0.6749
Into EM (MNI Atlas, freq=3)	0.1657	0.8970	0.9110	0.6579
Into EM (MNI Atlas, freq=10)	0.3933	0.8984	0.9226	0.7381
Using Label Propagation and Tissue Model				

A posteriori (our Atlas)	0.6454	0.8378	0.8655	0.7829
A posteriori (MNI Atlas)	0.4479	0.5900	0.7310	0.5897
Into EM (our Atlas, freq=5)	0.3963	0.8776	0.9035	0.7258
Into EM (our Atlas, freq=3)	0.3481	0.8770	0.9027	0.7093
Into EM (our Atlas, freq=10)	0.4872	0.9012	0.9267	0.7717
Into EM (MNI Atlas, freq=5)	0.2162	0.8970	0.9114	0.6749
Into EM (MNI Atlas, freq=3)	0.1657	0.8970	0.9110	0.6579
Into EM (MNI Atlas, freq=10)	0.3156	0.8967	0.9215	0.7113

Table 2. Summary of Results

4. Project Management details

In this project, the main tasks included designing an atlas-based segmentation framework, registering brain MRI volumes, and implementing label propagation. The initial setup, including selecting fixed images and configuring transformation models like rigid, affine, and B-spline, accounted for about 25% of the project time. Experimenting with different transformations to maximize segmentation accuracy required an additional 15%. Image registration was completed using Elastix and Transformix in Python and the command prompt, taking around 20% of the total time to test and refine. The label propagation phase involved aligning labels to the common reference space, with Transformix used in Python for efficiency, representing 15% of the project duration. Finally, atlas compilation, including template generation and probabilistic atlas creation, comprised about 25% of the project time. Overall, the project involved time distribution across setup, experimentation, and testing to achieve optimal segmentation results.

5. Conclusion

In this experiment, we successfully developed and evaluated an atlas-based segmentation framework for brain MRI tissue segmentation, focusing on white matter (WM), grey matter (GM), and cerebrospinal fluid (CSF). Our findings highlight the complexity of CSF segmentation due to intensity overlap with GM, making it challenging for intensity-based models alone to differentiate the two tissues effectively. By incorporating our atlas with both label propagation and an a posteriori approach, we significantly enhanced CSF segmentation results, underscoring the value of incorporating anatomical priors. Moreover, our atlas (a posteriori) approach consistently performed well across all tissue types, outperforming other methods and demonstrating robust results in segmenting WM, GM, and CSF. Overall, our atlas-based model has proven to be the most effective segmentation method across the main brain tissues, making it a reliable tool for medical image segmentation tasks.